Understanding Student Performance

Final Analytical Report | Scenario 2 | Mathematics

Research question: I wanted to find out which student characteristics are linked to their final Mathematics grades and their chances of passing. I also wanted to understand what we can and cannot say from these results.

1. Understanding and preparing the data

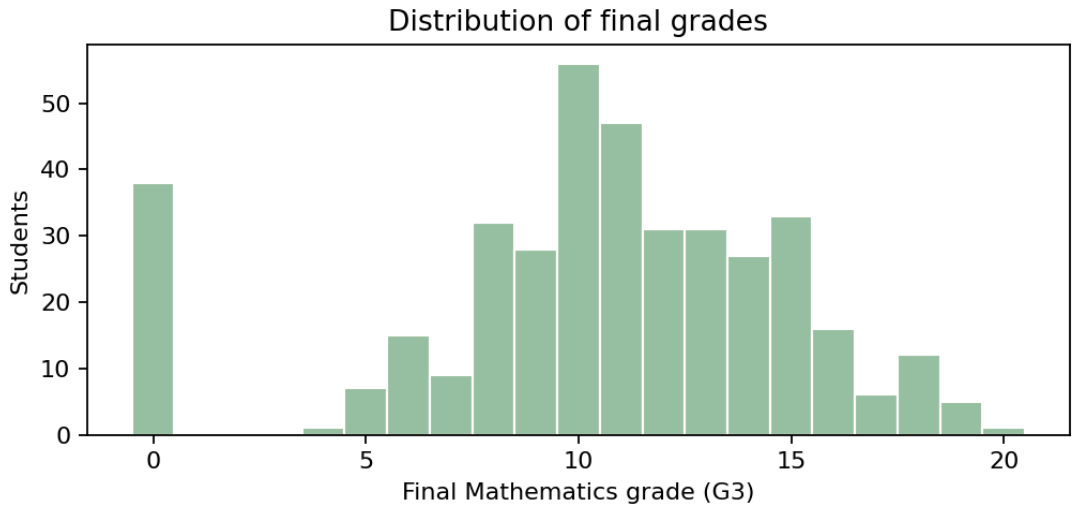
For this project, I used the UCI Mathematics student dataset. It includes 395 students from two secondary schools in Portugal, with 33 variables in total. Each row represents one student. The main outcome is their final grade (G3), from 0 to 20. I also created a pass/fail variable: students with a grade of 10 or above passed.

The dataset includes information such as age, absences, sex, study-time category, parents' education and whether students want to continue to higher education. Some variables are numbers, some are categories, and others are ordered categories. I kept the earlier grades G1 and G2 in the dataset but did not use them to predict G3 in the main models, as required.

I imported the file into R using **read.csv2()** because it uses semicolons. I checked the dimensions, variable types, missing values, duplicate rows and grade ranges. There were no missing cells or duplicate rows, and the final grades were within 0–20. Even so, the information might still contain reporting errors. Since these are student data, privacy matters, and I would avoid using the findings to judge individual students.

2. Exploring the data

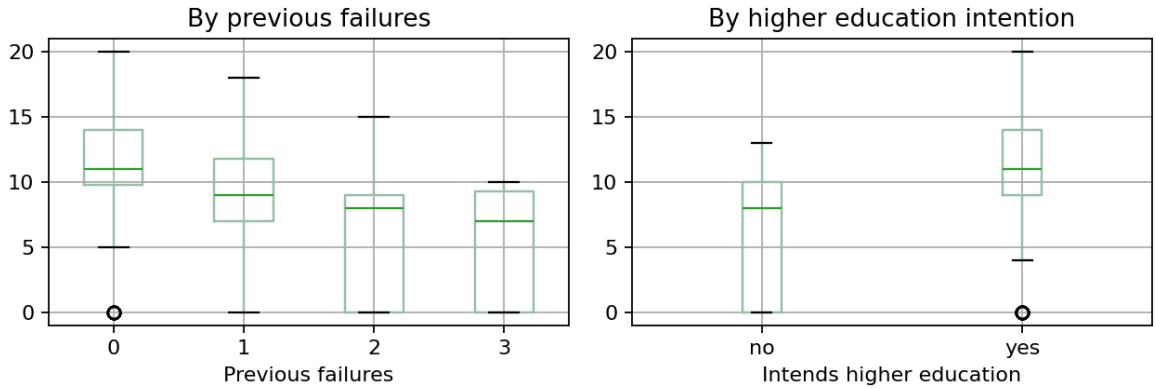
The average final grade was **10.42**, while the median was **11**. The standard deviation was **4.58**, and the middle half of grades ranged from 8 to 14. Overall, **265 students passed (67.1%)** and **130 failed (32.9%)**.



Exploring student grades

The histogram shows that many grades are around the passing mark, but there are also quite a few zero grades. This makes the distribution less balanced than a typical bell-shaped curve. The sample includes 208 female and 187 male students. Most students (375 out of 395) said they wanted to go on to higher education, while only 20 said they did not.

I also compared grades across different student groups. Students with more previous class failures generally had lower final grades. Students who wanted higher education tended to have higher grades, although the group saying “no” was much smaller. These patterns are useful starting points, but they do not tell us what caused the differences.



3. Looking at relationships

I used correlations to get a first idea of how some variables were related to the final grade. Previous failures had the clearest negative relationship with G3. Absences had almost no overall linear correlation in this dataset. I was careful with studytime because its values from 1 to 4 are categories, not exact hours of studying.

Variable	Correlation with final grade (G3)
Previous failures	-0.36
Absences	+0.03
Age	-0.16
Studytime	+0.10

Relationships and statistical models

I also compared pass rates by whether students wanted to continue to higher education. The difference was noticeable: 68.8% of those who said “yes” passed, compared with 35.0% of those who said “no”. However, there were only 20 students in the “no” group, so I would be cautious about making a broad conclusion.

Higher education intention	Passed	Failed
No	35.0%	65.0%
Yes	68.8%	31.2%

4. Building the regression models

I fitted two models using studytime, previous failures, absences, sex and higher-education intention. The **linear regression** looked at differences in final grades, while the **logistic regression** looked at the odds of passing. Both models left out G1 and G2. The reference groups were female students and students who answered “no” about higher education.

Predictor	Linear grade estimate	Pass odds ratio
Previous failures	-2.03	0.40
Male compared with female	+1.48	1.65
Higher education: yes	+2.06	2.18
Studytime	+0.42	1.12
Absences	+0.04	0.99

The strongest result was for previous failures. When the other predictors were held constant, one additional previous failure was linked to about **2.03 fewer grade points**. The odds ratio for passing was **0.40**, which suggests lower odds of passing. The model also estimated that male students scored about 1.48 points higher than female students after accounting for the included variables. This is a statistical comparison, not proof that sex causes a grade difference.

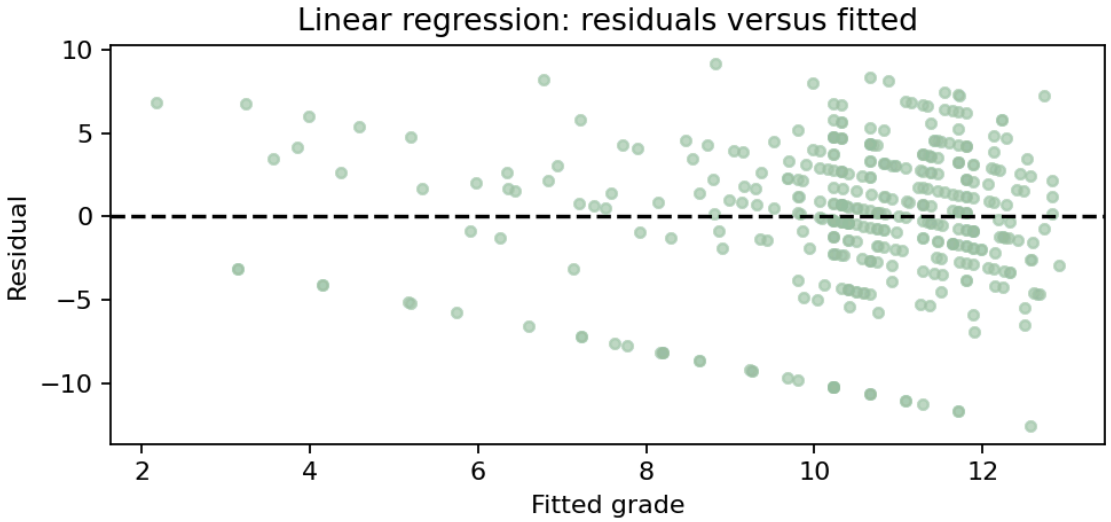
The linear model explained about **16.4% of the variation** in final grades ($R^2 = 0.164$). This means that most differences between students were still not explained by the variables I used. The logistic model added another perspective by looking specifically at passing rather than the exact grade. Its odds ratios should not be read as direct percentage changes in probability.

Model checks and uncertainty

5. How confident can I be in the results?

I checked confidence intervals and p-values instead of looking only at the size of the coefficients. Previous failures had a linear estimate of -2.03, with a 95% confidence interval from -2.62 to -1.43. Its logistic odds ratio was 0.40, with a 95% interval from 0.28 to 0.56. These results give fairly strong evidence of an association in this dataset.

For male compared with female students, the linear estimate was +1.48 (95% CI: +0.60 to +2.37), and the odds ratio was 1.65 (95% CI: 1.02 to 2.69). In comparison, studytime and absences had p-values above 0.05 in both models, so I would not describe their adjusted relationships as clearly supported here.



The residual plot shows quite a lot of variation, and the predicted grades cover a fairly narrow range. The zero grades may also affect the pattern. Together with the low R^2, this suggests the linear model misses many things that matter for student performance. Also, a statistically significant result is not automatically a large, useful or fair difference in practice.

Conclusions and reflection

6. What can I conclude?

The data are **observational**, meaning students were not randomly assigned to different conditions. Because of this, I can talk about relationships but cannot say that one characteristic directly causes a student to get a higher or lower grade. For example, students with previous failures tended to do worse, but that does not prove the failures themselves caused the later grades.

Other factors could help explain the results, such as previous academic ability, family circumstances, tutoring, motivation or teaching quality. Some information came from questionnaires, so answers might not always be completely accurate. Also, the students came from only two schools in Portugal, which means the results might be different in other countries or school systems. I did not test the logistic model on an independent dataset, so it should not be used to make automatic decisions about students.

What this could mean for schools

One useful takeaway is that students who have failed classes before may need extra academic attention. Schools could explore additional support or closer follow-up for these students. However, this project does not prove that a particular support programme would improve their grades. Any new programme should be checked carefully and should respect students' privacy.

My reflection

Doing this project helped me understand the difference between simply describing data and using regression to compare several factors at once. The hardest part for me was explaining the model results without making them sound like cause-and-effect relationships. If I had more time, I would test the models on new data, look at possible nonlinear relationships and compare the findings between schools.

Reference

Cortez, P. (2008). *Student Performance* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T

Reproducibility: The accompanying R script contains the analysis steps. The numerical findings in this report are based on the supplied Mathematics dataset.